

Introspection

- GraphQL introspection allows clients to query a server for its schema.
- Useful for tools like GraphQL Playground and Tyk Dashboard to understand available types and operations.
- Tyk uses introspection to fetch the upstream schema and render it in the UI when setting up a GraphQL proxy.

 Important: Introspection reflects the upstream schema only, not Tyk's schema transformations.

Introspection

Details

Upstream URL

https://countries.trevorblades.com/

☒ Upstream Protected

Upstream protected by

☒ Headers

☒ Certificate

☐ Persist headers for future use

Introspection headers

Key

Header name

Value

Header value

ADD

Domain

Add Certificate

SELECT CERTIFICATE

ENTER CERTIFICATE ID

Certificate

Select certificate

Introspection

GraphQL Introspection in Production

- Introspection is useful for development and debugging.
- In production, introspection can:
 - Reveal sensitive schema and implementation details.
 - Allow attackers to discover and map out your API.
- If Authentication Mode is Open (Keyless):
 - Introspection is always enabled and cannot be disabled.
- You can disable introspection per key or via security policy using:
 - Tyk Dashboard
 - Tyk Gateway API

Introspection Queries

- Any GraphQL API can be explored using introspection queries.
- Introspection queries reveal detailed schema information.
- Example: Introspecting all types in the schema.

```
query {  
  __schema {  
    types {  
      name  
      description  
      kind  
    }  
    queryType {  
      fields {  
        name  
        description  
      }  
    }  
  }  
}
```

Introspection Queries

- **Schema Overview Query**
 - Retrieves a summary of all types, queries, and mutations in the schema, including their names, kinds, and descriptions.
- **Single Type Details Query**
 - Fetches detailed information about a specific type, such as its fields, arguments, interfaces, enum values, and descriptions.
- **Field Arguments Query**
 - Focuses on the arguments of a specific field, detailing their names, types, and default values.
- **Enum Values Query**
 - Retrieves the possible values for an enum type, including descriptions and deprecation status.
- **Interface and Union Types Query**
 - Provides information on the possible types that implement an interface or belong to a union.